



Innovative 3D-Image Analysis of Cerebellar Vascularization Highlights Angiogenic Gene Dysregulations in a Murine Model of Apnea of Prematurity

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Abstract

Apnea of prematurity (AOP) affects 50% of preterm infants causing intermittent hypoxia (IH), which can lead to long-term neurodevelopmental deficits. Cerebellar abnormalities have been observed in AOP but the relationship between vascular alterations and neural development remains unclear. This study investigates how IH affects cerebellar angiogenesis using a murine model of AOP. We developed an innovative 3D imaging workflow combining Imaris and VesselVio software to quantitatively analyze cerebellar vascularization at different postnatal stages (P4, P8, P12, P21, and P70). We correlated these results with a transcriptomic analysis of 23 angiogenesis-related genes in the same stages to uncover the associated molecular pathways. We found that IH induced significant vascular changes, particularly at P4, with a global increase in vascular-network dimensions. By P8, the vascular network normalized, but genes were downregulated in all pathways studied. After P12, at the end of the IH protocol, transcriptional regulations had varied but persisted long-term. Moreover, differential analysis showed distinct effects on superficial versus deep vascular networks, allowing for a more precise understanding of remodeling patterns throughout development. Overall, transcriptomic changes were associated with morphological alterations in a time-dependent manner, suggesting a multiphasic IH response through development with lasting effects. Key regulations included VEGF, angiopoietin, and matrix metalloprotease signaling. These findings demonstrate that IH disrupts cerebellar angiogenesis in parallel with neurogenesis, potentially contributing to the neurodevelopmental deficits observed in AOP. Thus, the interconnected nature of angio- and neurogenesis during cerebellar development makes it crucial to take vascular aspects into account in therapeutic approaches to neurodevelopmental disorders.

Highlights

- Novel 3D imaging workflow reveals cerebellar vascular changes in apnea of prematurity mouse model.
- Intermittent hypoxia induces early hypervascularization at P4 followed by later adaptation.
- Superficial and deep vascular networks show distinct responses to intermittent hypoxia.
- Vascularization-related genes are affected during cerebellar development.
- Alterations in angiogenesis go together with neurodevelopmental defects.

Keywords Vascularization · 3D imaging · Cerebellum · Apnea of prematurity · Neurodevelopment · Transcriptomics

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Introduction

Apnea of prematurity (AOP) consists of an abnormal respiratory rhythm where breathing is interrupted at least every 5 min and for at least 20 s, thus inducing a state of intermittent hypoxia (IH). Premature newborns are particularly prone to AOP due to their immature respiratory system [1, 2]. This condition affects 50% of all preterm infants and nearly 100% of neonates born under 28 gestational weeks. The neonates will typically resume a normal breathing pattern by the corrected term, but several metadata provide evidence of a correlation between the duration of AOP and the incidence of developmental abnormalities that lead to long term behavioral deficits [3–5]. Among the reported deficits, children suffer from motor coordination, language, and spatial orientation impairments, suggesting that a cerebellar alteration could be involved. According to this hypothesis, we have recently demonstrated, thanks to an established mouse model of AOP [6], that IH induces a delay of cerebellar cortex maturation and an alteration of the dendritic arborization of Purkinje cells, with long lasting transcriptional alterations [7, 8]. The strong effect of AOP on the cerebellum is due to the developmental window of this nervous structure which mainly occurs after birth.

Indeed, in mice as in humans [9, 10], cerebellar cortical development starts during embryogenesis from two germinative regions, the rhombic lip and the ventricular zone. Glutamatergic granule cells come from the rhombic lip and migrate tangentially to form the transitory external granular layer (EGL) at the surface of the cerebellum. GABAergic Purkinje cells (and several types of interneurons) arise from the ventricular zone and migrate transversally to the surface to create a monolayer of Purkinje cells just below the EGL. As a result, at birth, the cerebellum is composed of 3 layers, the EGL, containing granule cell precursors (GCP) and the Purkinje cell layer (PCL), separated by the molecular layer (ML). However, at this point, the structure is still immature and has yet to undergo postnatal maturation. Firstly, the GCP proliferate intensively and cross the ML and the PCL to form the definitive granule cell layer (GCL) until the total disappearance of the EGL. Meanwhile, the GCP finish their maturation and send their axons, called parallel fibers, to the ML. At the same time, Purkinje cells develop their dendritic trees into the ML to contact parallel fibers [11]. Much like in the rest of the brain, the neurohistogenetic process is mirrored by angiogenesis within the cerebellar cortex. Then the cerebellar vascularization initiates in the perineural vascular plexus that covers the neural tube [12]. Then, collaterals from pial vessels enter the cerebellar cortex and irrigate the EGL at first sparse in the EGL during the proliferative phase of GCP, blood vessels then continue branching out and the

number of capillaries increases in the ML and the GCL [13]. Therefore, as hypoxia is known to remodel the vascular organization and enhance the density of blood vessels in the brain [14, 15], we can hypothesize that AOP could disrupt vascularization during the postnatal development of the cerebellum (as it does with neuronal cells).

In line with this hypothesis, our previous transcriptomic results showed that the expression of the hypoxia inducible factor 1 alpha (HIF1 α) is affected in response to our IH protocol [7]. HIF1 α is considered as the main regulator of angiogenesis since this factor responds to low levels of dioxygen (O₂) by promoting the expression of pro-angiogenic actors, including the vascular endothelial growth factor (VEGF) and platelet derived growth factor β (PDGF β) as well as miRNA [16]. Another angiogenic process could also occur in the brain via the Ang2/Tie2 pathway (for *Angiopoietin2/tyrosine kinase containing immunoglobulin and epidermal growth factor homology domain-1*). Indeed, under hypoxic conditions, endothelial cells increase Ang2 production which acts in an autocrine manner on Tie2 receptor to destabilize capillaries and promote angiogenesis [17]. Interestingly, Tie2 has been recently described in Purkinje cells and it has been demonstrated that the neuro-vascular signalling Ang2/Tie2 controls dendritic morphogenesis of Purkinje cells during cerebellar postnatal development [18].

All these data led us to consider that the alteration of the Purkinje dendritic trees observed after IH in our AOP model could be linked to a perturbation of cerebellar vascularization. To provide elements in understanding this mechanism, we performed a transcriptomic study focused on the main factors involved in cerebellar angiogenesis at different postnatal stages. Moreover, to correlate putative modifications of gene expression with defects in vessel morphology, we developed an image analysis workflow using both Imaris and VesselVio software to visualize the cerebellar vascularization in 3D during postnatal development and obtain comparable quantitative parameters between normoxic and IH conditions.

Materials and Methods

Animals

Animals used in this study were wild type C57Bl6/J mice born and bred in an accredited animal facility (approval number A 76–451–05). Animal experiments were approved by the French Regional Ethics Committees and the Ministry of Education, Research and Innovation (project n°10310-2017041811326427v5–13/02/19–12/02/22), and they were performed in accordance with the European Committee

Council Directive (2010-63-EU). Mice lived on a 12-hour light/dark cycle and had free access to food and water. Sex was determined based on anogenital distance measurement as well as pigment-spot localization [19]. From the postnatal day 2 (P2) onwards, mice were assigned a unique identifier before initializing the IH protocol. There was no blinding in this study, and sample size was chosen based on power determination from our preliminary studies [7, 8]. A total of 28 litters of animals including their dams were used for this study.

Intermittent Hypoxia Protocol

Our IH protocol relies on a custom hypoxia chamber, which is based on the protocol developed by Cai et al. [20]. Briefly, IH was achieved by the repeated succession of 2-min cycles of hypoxia (5% O₂; 20 s/cycle) and reoxygenation, for 6 h per day, at 21±0.5 °C, while mice were in their sleep phase (10 am–4 pm). The protocol was initiated on neonatal P2 pups (IH group) until the desired stage, or for 10 days maximum. In terms of cerebellar neurodevelopment, several studies have shown that the period between P2 and P12 in mice corresponds to a developmental stage ranging from prematurity to normal gestational age birth in humans [21, 22]. Moreover, exposure to these hypoxic cycles induces deficits similar to those observed in preterm infants with AOP, such as intermittent desaturations, bradycardia and hypomyelination [6–8, 20, 23, 24]. These findings therefore support the validity of this protocol as a murine model of this pathology. Throughout the protocol, the following parameters were constantly monitored in the chamber: oxygen concentration, hygrometry, temperature, and atmospheric pressure. The control/normoxia group (N) was placed in a chamber open to room air but mimicking the hypoxia chamber's environment to control for external stressors.

Sample Gathering

For the transcriptomic study, mice were sacrificed at stages P4, P8, P12, P21, and P70 by decapitation after anesthesia by isoflurane inhalation (Iso-VET). Whole brains were immediately harvested and set in pure isopentane at −30 °C. They were then stored in sterile containers at −80 °C until further use. Sample sizes for RT-qPCR were: 13 for P4, of which 7 N (3 Females (F)+4 Males (M)) and 6 IH (4 F+2 M); 16 for P8, of which 6 N (3 F+3 M) and 10 IH (6 F+4 M); 25 for P12, of which 15 N (8 F+7 M) and 10 IH (2 F+8 M); 17 for P21, of which 9 N (3 F+6 M) and 8 IH (5 F+3 M); and 8 for P70, of which 5 N (1 F+4 M) and 3 IH (1 F+2 M).

For imaging studies, brains from P4 mice were directly harvested and fixed by immersion in 4% paraformaldehyde, given their small size. For P8, P12, P21, P70 stages, mice were lethally anesthetized by intraperitoneal injection of ketamine (100 mg/kg) and xylazine (10 mg/kg), and then sacrificed by intracardiac perfusion of NaCl 9‰ and paraformaldehyde 4%, before removing the brains. Brains were then postfixed overnight in 4% paraformaldehyde, and stored in phosphate buffer saline (PBS). Sample sizes for the clearing experiments were: 8 for P4, of which 4 N (2 F+2 M) and 4 IH (2 F+2 M); 8 for P8, of which 4 N (2 F+2 M) and 4 IH (1 F+3 M); 7 for P12, of which 3 N (1 F+2 M) and 4 IH (1 F+3 M); and 7 for P21, of which 3 N (2 F+1 M) and 4 IH (0 F+4 M).

Panels and Primer Design

The vascularization gene panel was built based on an analysis conducted with the Cytoscape software (v.3.9.1; [25]) using protein interaction data retrieved from STRING-DB [26]. The panel was further expanded by enrichment with the plugin stringApp (v 2.0.1; settings: maximum additional interactors=10, confidence cutoff=0.40; [27]) and then cross referenced with ClueGO pathway (v.2.5.9; [28]) and functional data retrieved from the literature. For readability and ease of interpretation, these findings are summarized in Table 1, and the resulting functions were used to group the quantitative RT-PCR results.

Gene primers were designed with the Primer Express software (v3.0.1; ThermoFisher Scientific) using nucleotide sequences from the NCBI Pubmed database. Primer pairs were ordered from Integrated DNA Technologies and their specificity was validated by linear regression of serial dilution data. Primer pairs were chosen preferentially to be on exon joining sites, with the least possible hairpin and dimer formation, and with similar size, GC percentage, and melting temperature for forward and reverse primers. Each sequence was blasted on Basic Local Alignment Search Tool (Blast [29]), to ensure specificity. See Supplementary Tables 1 and 2.

RNA Extraction

The samples were homogenized in 1 ml of Trizol (ThermoFisher) and mRNAs were purified on column using the Nucleospin RNA extract II from Macherey-Nagel (cat. 740 955 250) according to manufacturer recommendations. RNA quantity and purity were analyzed by UV spectrophotometry (Nanodrop Technologies). The optical density (OD) of RNA was read at 230, 260, and 280 nm. The ratios

Table 1 Selected angiogenesis-associated genes, the functions they are associated with, and their roles within these functions. Angpt1: angiopoietin 1; Angpt2: angiopoietin 2; Anpep: alanyl aminopeptidase; Cdh5: vascular epithelium-cadherin; Colla1: collagen, type I, alpha 1; ECM: extracellular matrix; F3: coagulation factor III; Fgf2: fibroblast growth factor 2; Flk1: vascular endothelial growth factor receptor 2; Flt1: vascular endothelial growth factor receptor 1; Lect1: chondromodulin; Mmp2: matrix metalloproteinase 2; Mmp9: matrix metalloproteinase 9; Ndp: Norrie disease protein; Nrpl: neuropilin 1; Pgf: placental growth factor; Serpine1: plasminogen activator inhibitor 1; Serpinf1: pigment epithelium-derived factor; Tek: endothelial-specific receptor tyrosine kinase; Tgfb1: transforming growth factor, beta 1; Thbs1: thrombospondin 1; Tie1: tyrosine kinase with immunoglobulin-like and EGF-like domains 1; Timp1: tissue inhibitor of metalloproteinase 1; Vegfa: vascular endothelial growth factor A

Function abbreviation	Functional pathway	Genes with a putative positive effect on the function	Genes with a putative negative effect on the function
VSM	Vascular stabilization and maturation	<i>Tgfb1; Flk1; Angpt1; Tek; Vegfa</i>	<i>Tie1</i>
EEP	ECM and endothelial permeability	<i>Cdh5; Vegfa; Mmp2; Mmp9; Angpt2</i>	<i>Angpt1</i>
VIS	Vascular invasion and sprouting	<i>Cdh5; Fgf2; Vegfa; Serpine1; Nrpl; Anpep; Mmp9; Ndp; F3</i>	<i>Flt1; Colla1; Thbs1; Timp1</i>
EPS	Endothelial proliferation and survival	<i>Pgf; Tgfb1; Flk1; Nrpl; Angpt1; Tie1</i>	<i>Lect1; Angpt2; Thbs1; Serpinf1</i>
VRP	Vascular remodelling and patterning	<i>Flt1; Tie1</i>	<i>Vegfa</i>
HIA	Hypoxia-induced angiogenesis	<i>Mmp2; Mmp9; Tek; Tie1; Vegfa; Serpine1</i>	<i>Serpinf1</i>

OD 260 nm/OD 280 nm and OD 260 nm/OD 230 nm were calculated as indicators of protein, and salt/ethanol contamination, respectively. Ratio values between 1.6 and 2.0 were considered acceptable. As per the MIQE guidelines [30], mRNA quality assessment was performed by a bio-analyser gel electrophoresis on RNA 6000 Pico chips (cat. 5067–1513, Agilent), and samples with an RNA integrity number (RIN) between 7 and 10 were considered qualitative enough to be analyzed. The mRNAs were then stored at -80°C until use.

Quantitative RT-PCR

Messenger RNAs were converted to cDNA by reverse transcription using the Prime Script RT reagent kit (cat. RR037A, Takara). Relative gene expression level determination was done by real time PCR in 384-well plates. Total reaction volume was 5 μL including: 2.5 μL 2X Fast SYBR Green PCR Mastermix (cat. 4385612, Thermofisher), target

gene-specific sense and antisense primers (0.15 μL of each, 100 nM final concentration), 1 μL PCR-grade water, and 1.2 μL of sample solution. The cDNA samples and reaction mixes were distributed via the Bravo 1 liquid handling platform (Agilent). The real time PCR reaction took place in a QuantStudio Flex 12k thermal cycler (Applied Biosystems). For each gene of interest (GOI), sample measurements were conducted at least in duplicate and with a minimum of two housekeeping genes (HKG). Values were calculated via the $2^{(-\Delta\Delta C_q)}$ method where:

$$2^{-\Delta\Delta C_q} = 2^{(-(C_{qGOI_{IH}} - C_{qHKG_{IH}}) - (C_{qGOI_N} - C_{qHKG_N}))}$$

Imaging

Cerebella previously fixed with 4% paraformaldehyde underwent a clearing protocol consisting of the following steps. Samples were dehydrated by being submerged in increasing concentrations of methanol (MeOH; 20%, 40%, 60%, 80% and 100%). Dehydrated samples underwent bleaching in a solution of 5% H_2O_2 (hydrogen peroxide) and 95% MeOH for 24 h in order to decrease tissue autofluorescence. Samples were then rehydrated by being submerged in decreasing concentrations of MeOH (80%, 60%, 40% and 20%), then washed in solution PTx.2 (100 mL PBS 10X+2 mL Triton-X100+Q.S. 1 L distilled water). Samples were then permeabilized for one day at 37°C under agitation with a solution containing 1X PBS, 0.2% TritonX-100, 20% dimethyl sulfoxide (DMSO), glycine (23 mg/mL) and thimerosal at 0.1 g/L (antifungal). Finally, non-specific binding sites were blocked with a 1X PBS solution containing 0.2% Triton-X100, 10% DMSO, 6% normal donkey serum (NDS) and thimerosal for one day at 37°C under agitation.

Thereafter, the brains were incubated with the primary antibodies diluted in PTwH solution (100 mL PBS 10X+200 μL Heparin (50 mg/mL)+Q.S. 1 L distilled water), containing 5% DMSO and 3% NDS at 37°C under agitation for 6 days. After 6 rinses with PTwH at room temperature, the samples were incubated for 5 days with the appropriate secondary antibodies (Table 2) diluted in PTwH containing 3% NDS at 37°C under agitation. The brains were then rinsed several times with PTwH at room temperature under agitation, followed by dehydration in MeOH baths of increasing concentration (20%, 40%, 60%, 80% and 100%). A delipidation of the brains was then performed by incubation in a solution containing 66% dichloromethane (DCM) and 33% MeOH for one night under agitation and then in DCM 100% for 30 min. These last two steps homogenize the refractive indexes of the cellular structures and induce their transparency once placed in dibenzylether.

Table 2 Antibodies used for the visualization of blood vessels. α -SMA: smooth muscle actin alpha; DAG: donkey anti-goat; DART: donkey anti-rat; N/A: not applicable; PECAM1: platelet endothelial cell adhesion molecule 1. The secondary antibodies were purchased from Molecular Probes

Primary Antibodies	Target	Dilution	Species	Supplier	Secondary antibodies	Dilution
Podocalyxin	Capillaries	1:200	Goat	R&D Systems (#AF1556)	DAG-Alexa 594	1:400
α-SMA-Cy3	Arteries	1:500	Mouse	Sigma-Aldrich (#C6198)	N/A	N/A
PECAM1	Arterioles	1:200	Rat	Millipore (#CBL1337)	DART-Alexa 594	1:400

The 3D acquisitions of the transparent cerebella were performed on the Ultramicroscope Blaze (Miltenyi Biotec) using the software InspectorPro (version 8.0). Image analysis and vessel modelling were done using the Imaris software (Oxford Instruments, version 10.0) and then transferred to VesselVio (version 1.2) software to obtain quantitative and comparative parameters [31].

Statistical Analysis

Statistical analyses were performed within the R statistical computing environment (version 4.5). Both real time PCR and imaging data were modeled through the Generalized Linear Mixed Model (GLMM) framework, using the {glm-mTMB} package [32]. For real time PCR data, a Gaussian likelihood with an identity link function was used to model the distribution of the DCq samples for each gene of interest. When DCq samples for one gene were split over multiple plates, a random intercept was added to account for intra-plate correlations. Additionally, in order to explore the multivariate structure of the morphometric parameters of vascularization, a principal component analysis (PCA) was performed using the {FactoMineR} package [33], after centering and scaling the data. PCA results were visualized with the {factoextra} package with individual points colored according to experimental group [34]. Convex ellipses were added to visualize the dispersion of groups in the principal component space.

Model diagnostics were done using the {DHARMA} [35] and {performance} [36] packages. The fitness of each model was assessed through both visual checks (e.g., posterior predictive checks, QQ plots, and residuals vs. predicted values) and quantitative indices of model fit (e.g., AIC: Akaike Information Criterion). When several competing models were possible a priori, we selected the most plausible one primarily based on our theoretical understanding of the response's properties, and, to a lesser extent, to minimize AIC and favor model parsimony.

Contrasts and p-values for relevant hypotheses were obtained using the {emmeans} package [37]. They were computed on the link scale, using Wald t-tests, without any multiplicity adjustments. For all analyses, $p < 0.05$ was considered significant and for each figure asterisks indicate the level of statistical significance: one for $p < 0.05$, two for $p < 0.01$, and three for $p < 0.001$.

Results

Workflow of the Cerebellar Vascularization Analysis

Thanks to a concomitant triple labelling (smooth muscle actin alpha (α -SMA), platelet endothelial cell adhesion molecule 1 (PECAM1) and podocalyxin, we were able to visualize all the vascular networks, namely, the arteries, veins, and capillaries, at all stages of postnatal development (Fig. 1). The labelling is present in all cerebellar structures. Signal intensity is stronger in the cortex than in the depth of the tissue at P12 and P21 because of the higher volume of the sample that impedes antibody penetration despite applying an increasing permeabilization protocol. Due to the variability of the cerebellum's morphology, no method currently exists to analyze the vasculature of postnatal mice cerebella. Therefore, we developed the following workflow by optimizing each step of analysis using Imaris and Vesselvio software.

The file resulting from the 3D lightsheet acquisition was converted to an .ims format, then the *Surface* function of the Imaris software was used to delineate the cerebellum from the whole hindbrain 3D acquisition (Fig. 2A). Because of the complexity of the cerebellum's shape, the delineation was performed manually, by drawing cerebellar outlines every 20 μ m in the 2D-slice mode (Fig. 2B). The resulting *Surface* was used to obtain a channel (mask) including only the cerebellum (Fig. 2C), which allows the calculation of the total cerebellar volume as well as the total volume of the vasculature as follows. The vascularization modelling was built with the *Filament* modality and applied to the cerebellum channel. The *Automated Autopath Algorithm* mode was chosen among the six algorithms proposed by the software because this process models dense networks with numerous branches. The algorithm is comprised of three steps and is primed on a representative region of interest.

The first step involves choosing the detection modalities. The observation of the vascular organization on a sagittal cerebellar section revealed a high heterogeneity of vessel diameter (Fig. 2D). Large vessels are located on the surface of the cortex, and fine vessels are in the center of the cerebellum (Fig. 2E). Thus, to optimize the quality of the modelling beforehand, we split the vascularization into two networks based on a threshold of voxel intensity. All the positive voxels were considered as the "superficial network" and all

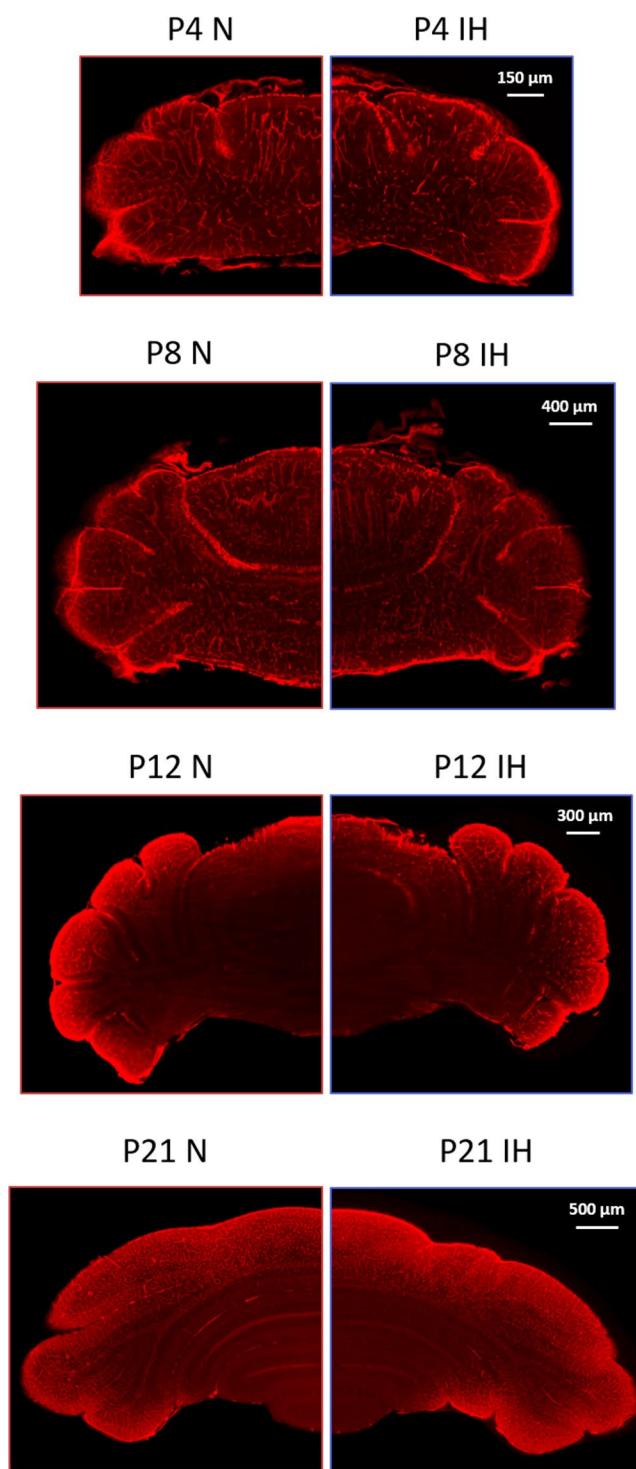


Fig. 1 Images illustrating the cerebellar vascularization in normoxia and intermittent hypoxia conditions. Lightsheet microscopy 2D images of cerebella illustrating the vascularization in control (N) and hypoxic mice (IH) at various postnatal stages (P4, P8, P12, and P21). Whole cerebella were simultaneously incubated with antibodies against PECAM1, podocalyxin and α -SMA to label all blood vessel types before clearing and image acquisition. For each stage, the scale is indicated in the top right corner. IH: intermittent hypoxia; N: normoxia; Px: postnatal day x

Fig. 2 Images illustrating the Imaris workflow developed for the vascular network modeling. **A–I**: Sequential workflow steps allowing the analysis of the cerebellar vascular network of a P4 mouse cerebellum on the Imaris software. From a 3D lightsheet acquisition (**A**), the cerebellum is delineated (**B**) and a mask is created (**C**). Within that selected volume, the cerebellar vascularization is segmented (**D**), which allows the network visualization (**E**) and the separation of a deep and a superficial network (**F**). Then the threshold of seedpoints is defined (**G**), and thanks to the artificial intelligence module (AI), Imaris is able to discriminate “true” (blue) and “false” (red) seedpoints (**H**), and “true” (blue) and “false” (red) segments (**I**). AI: artificial intelligence; Px: postnatal day x

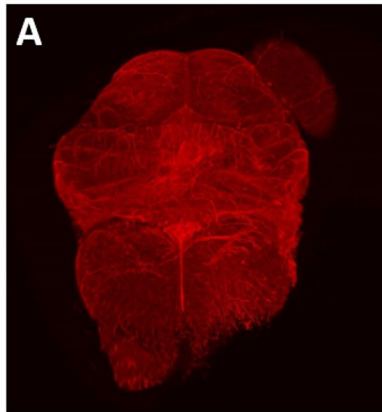
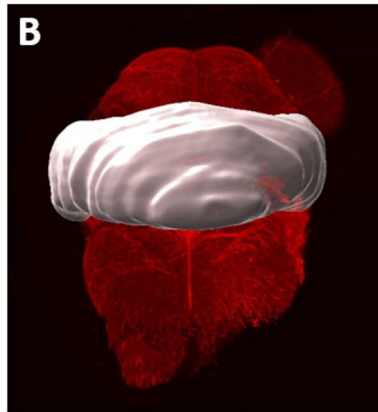
the negative voxels represented the “deep network”, which includes all the vessels in the organ’s core (Fig. 2F).

The second step of the protocol involves defining the diameter range to be considered. These diameters constitute the detection threshold for branching points, defined as the junctions between each vascular segment. The network splitting allowed more homogeneity of the vessel diameters. However, we chose a *Multiscale* option to guarantee the detection of all vessels. Thus, a diameter range of [1–10 μ m] was chosen for the “deep network” and a [1–20 μ m] range for the “superficial network.”

The third step of the algorithm requires training the artificial intelligence (AI) in three different phases. In the first phase, the aim is to define the threshold for detecting orientation changes. This step is crucial because it enables the subsequent training of the software to classify these points and then generate the segments. So, the wider the range is, the more efficient the training will be (Fig. 2G). In the second phase, based on the threshold, the AI predicts keeping or discarding seedpoints. The AI can be further trained by correcting potential choice errors until a satisfactory result is achieved, but the correction must not exceed 100 manual points to avoid conflicting information, which could disrupt the training (Fig. 2H). With the same method, the third phase consists of classifying and selecting the resulting segments (Fig. 2I). Finally, the process is applied to the entire 3D volume (Fig. 3A). To ensure reproducibility, training was first performed at P4 on 20 ROI from two separate cerebella. Once optimized, the parameters were applied to all normoxic and hypoxic cerebella of the same age without any modification. This process applied at P4 then served as the basis for optimization at P8, then that of P8 for optimization at P12, and so on. The main Imaris steps of vessel modeling are illustrated in the supplementary data (supplementary Video 1).

Then, for each network (deep and superficial), a channel was created from the filament modelling via a Matlab macro linked to Imaris (*Image Processing Create channel from Filaments*) and then opened in the Fiji software via the *Image to Fiji* bridge. Thus, the files were converted in 8-bit images, binarized, and saved into *.tiff* and *.nii* formats. Each file is then opened in the VesselVio software and the

3D acquisition

Cerebellum
delineation

Cerebellum mask

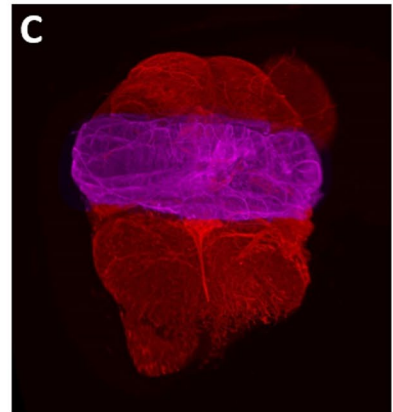
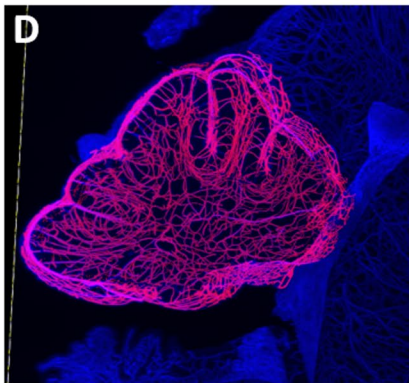
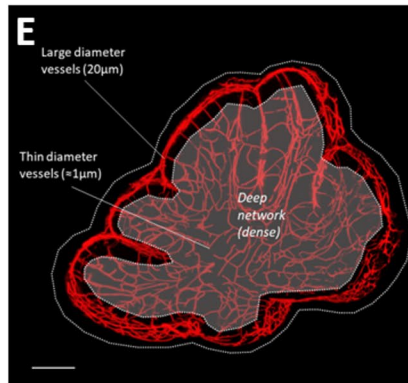
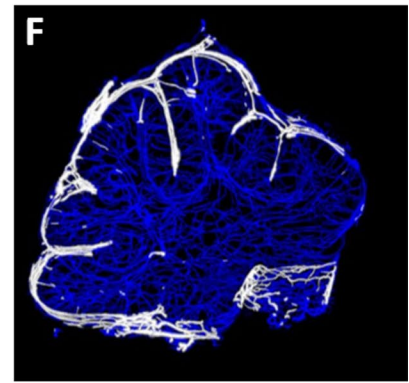
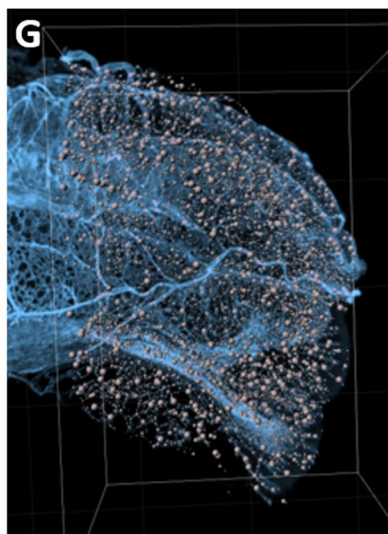
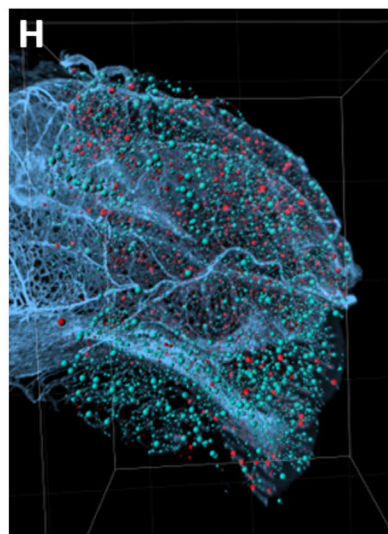
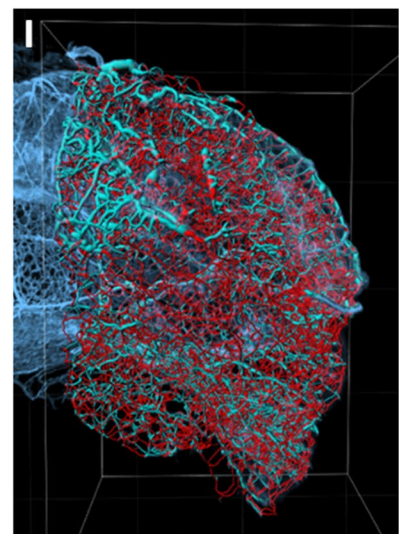
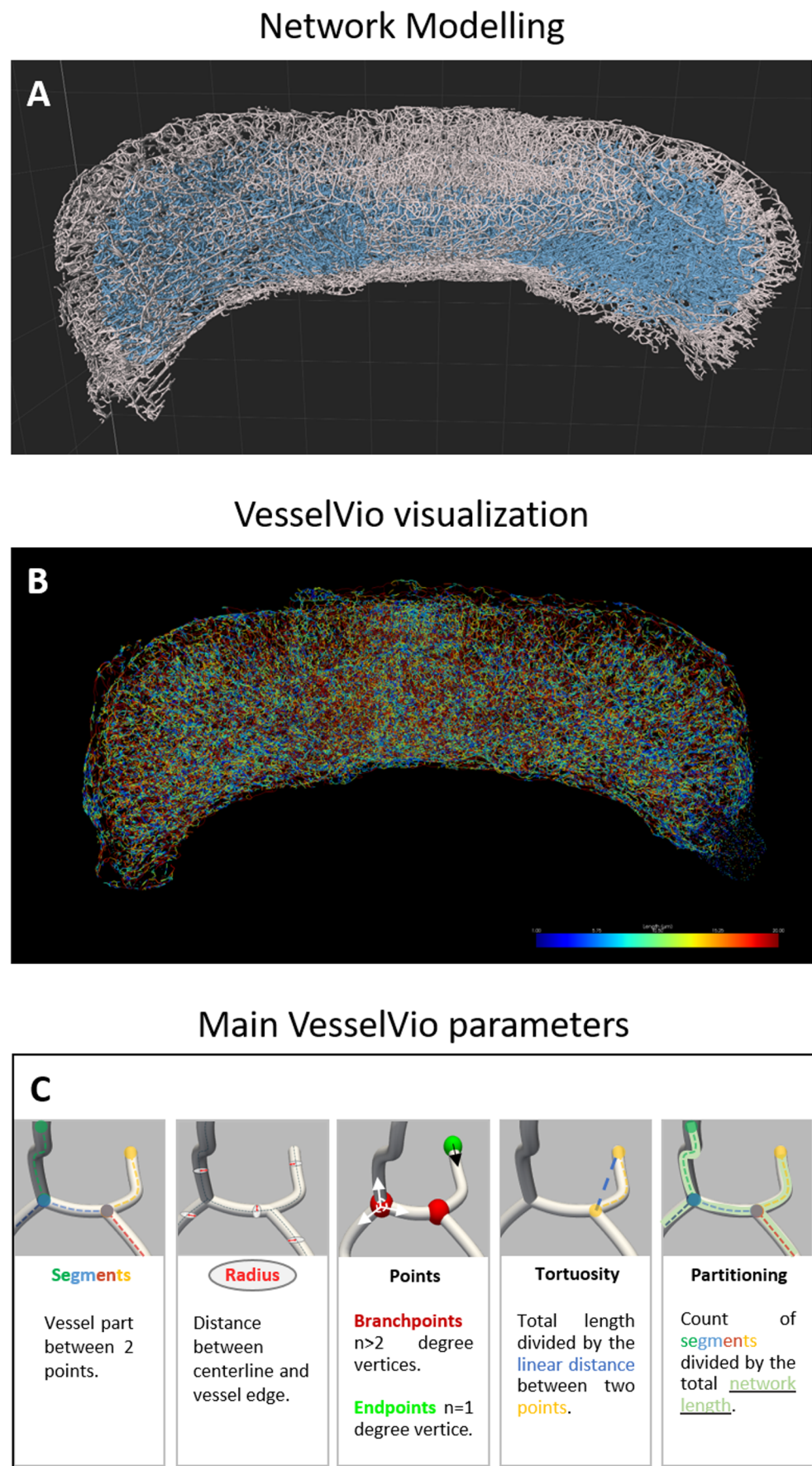
Cerebellum
vascularisationNetwork
parametersMasks of the 2
networksSeedpoint
thresholdingSeedpoint
AI learningSegment
AI learning

Fig. 3 Illustration of the vascular network modeling of a postnatal mouse cerebellum and the parameters defined on VesselVio software. **A, B:** Illustration of the matching between the Imaris network modelling (**A**) and the corresponding VesselVio visualization (**B**). In **A**, the superficial network is represented in white and the deep network is represented in blue. In **B**, the different ranges of segment length are indicated by the colored scale. **C:** Graphical definition of the network parameters analyzed with VesselVio



vascular skeletons were visualized in *.tiff* files to check file conversion before analyzing (Fig. 3B). Then, all segmented vasculature datasets were downloaded in *.nii* files for analysis. Finally, a single excel file is created with the specific parameters of each vascular network. In this study, the parameters of interest were (i) the total volume, area, length and number of segments, for the whole cerebellar network, (ii) some specific vessel characteristics such as the branchpoints, endpoints, tortuosity and segment partitioning, and (iii) the mean volume, area, length and radius, for segments (Fig. 3C).

Physiological Evolution of the Vascular Network During Cerebellar Postnatal Development

Because of the notable inter-individual variability, all the following results were normalized relative to the total volume of the cerebellum.

Based on all the parameters studied, it is clear that the vascular network undergoes drastic changes during postnatal development of the cerebellum. At P4, the length and volume of the network are highest relative to the size of the cerebellum. These parameters decrease until P12 before increasing again at P21 (Fig. 4A). The same is true for the number of segments and branches, indicating a remodeling of the vessels over time. This remodeling is observed in 3D models created on a 55- μ m thickness section of lobule VI at different stages of development (Fig. 4B). At P4, large-diameter vessels penetrate the cortex and delimit the lobule. Smaller caliber vessels depart transversely, cross the EGL and the ML to join the IGL. In this layer, a dense network is formed by penetrating vessels originating from the superficial network. At this stage, the vascular network reveals a specific organization depending on the layer considered. At P8, vessels parallel to the axis of the convolution appear in the cortex, revealing a boundary between the EGL and the ML. At P12, this boundary becomes blurred, but numerous penetrating vessels continue to develop from the superficial network, allowing significant vascularization of the IGL and lobular white matter. Finally, at P21, the vessels delimiting the lobule are still identifiable with a few large-diameter collaterals, but the density of the capillaries is such that it is difficult to distinguish the cortex from the white matter.

Effect of IH on the Vascularization of the Postnatal Cerebellum

PCA analyses reveal that vascular parameters differ significantly between normoxic and IH animals, with the effect of IH being most pronounced at P4 and diminishing over time, as the two groups tend to overlap from P12 onwards (Figs. 5A, 6A, 7A and 8A). Concerning the general parameters of

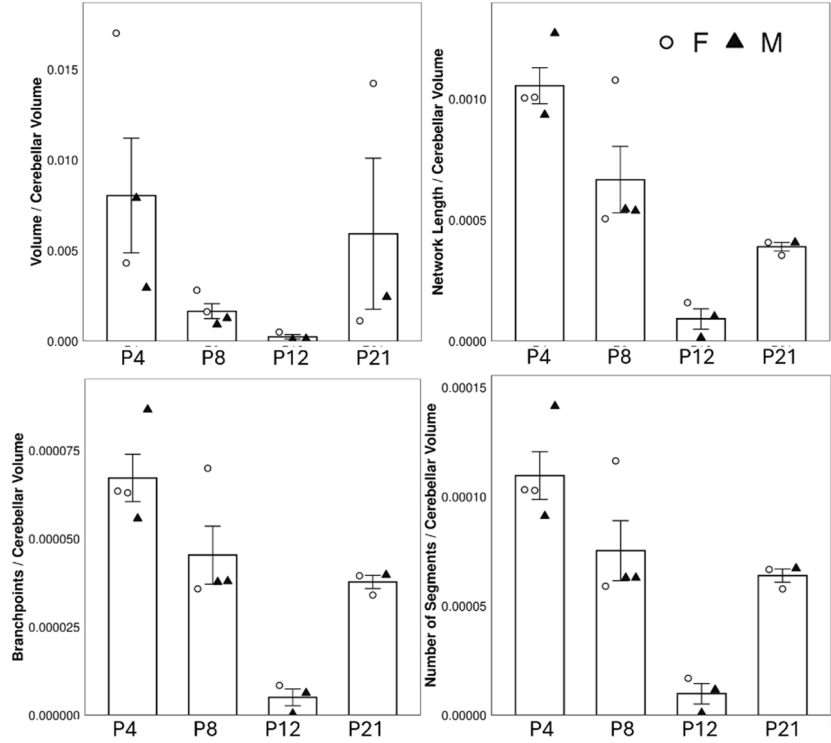
the entire vascular network, the IH protocol induced some major modifications at P4 with a strong increase in the volume, area and length, indicating a denser network from the very first days (Fig. 5B). The higher number of segments, branchpoints and partitioning confirms the increased complexity of the vascularization after hypoxia. In contrast, the tortuosity is decreased at P4 in hypoxic animals, indicating an inhibiting effect on vessel remodeling (Fig. 5C). Thanks to our analysis workflow (Fig. 2F), that makes a differential analysis of the superficial and deep networks possible, we showed that the differences observed for the volume, the length and the partitioning are mainly due to an effect of IH on the superficial network, whereas the increase of the segments and branchpoints is attributable to the deep vessels (Fig. 5C). From P8 onwards, all the general IH-induced modifications are attenuated and become mostly non-significant (Figs. 6, 7 and 8), suggesting that hypoxia no longer has an effect on the cerebellar vasculature at later stages. However, our differential study revealed network dependant effects at P8 and P21. Thus, IH induced a specific increase of the deep vessel partitioning at P8 (Fig. 6B) and a specific higher tortuosity of the superficial vascular segments at P21 (Fig. 8B). The number of endpoints is also affected by IH at P21 (Fig. 8B) but, by looking at the two networks separately, it appears that this decrease is also present at P8 and mainly affects the deep vessels (Fig. 6B).

Meanwhile, the effects of IH on segment characteristics are more persistent and variable between ages. At P4, all mean segment parameters are affected by IH with an increase of the mean volume but a concomitant decrease of the area, length and radius (Fig. 5D). All these variations are mainly due to the superficial network, whose tortuosity is also specifically altered (Fig. 5D). At P8, we only observed a persistent higher mean volume of segments after IH in studying both networks together (Fig. 6C) but the differential analysis showed a decreased mean length of deep segments and an increased mean area of superficial segments, which are cancelled by a non-significant opposite effect of IH on the deep network (Fig. 6C). At P12, hypoxia induced a decrease of the mean segment area largely attributable to the superficial vessels, whereas IH reduced the mean segment volume of both networks (Fig. 7C). Finally, at P21, 6 days after the end of the IH protocol, no effects are observed on the mean segment parameters (Fig. 8).

Effect of IH on the Expression of Angiogenesis-Related Factors

The transcriptomic analysis was targeted at genes known to be involved in angiogenesis and performed on whole cerebella at different stages of postnatal development (Fig. 9). We tested a panel of 23 genes, of which 21 were

A



B

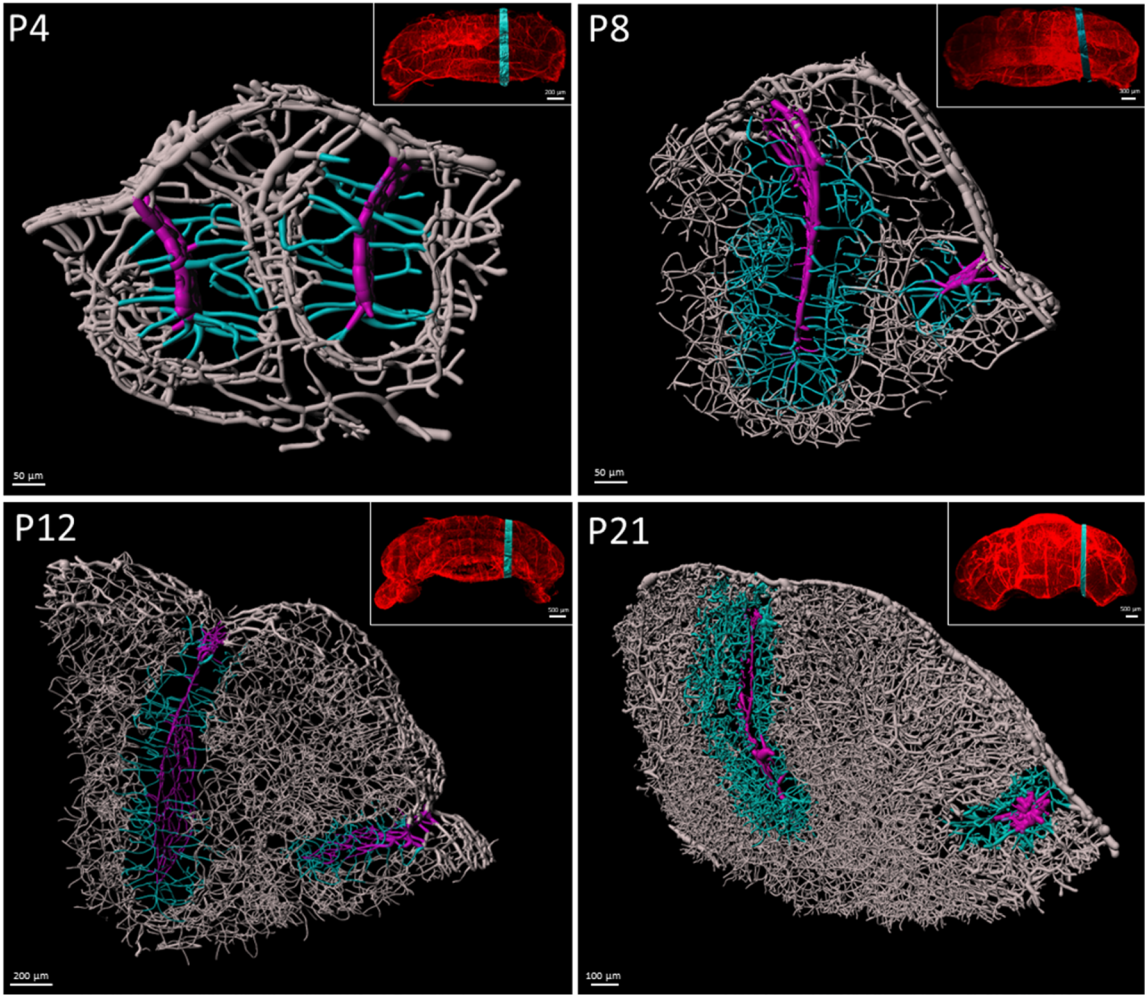


Fig. 4 (next page) Illustration of the evolution of the cerebellar vascularization during the physiological postnatal development in mouse. **A:** Measurements of volume, length, branching points, and the number of segments in the cerebellar vascular network in mice at postnatal days P4, P8, P12, and P21, illustrating the changes in key vascular parameters during postnatal cerebellar development. Each value represents the mean $\pm$ SEM of 4 (P4 and P8) or 3 (P12 and P21) animals, among which females are indicated by a circle and males by a triangle. **B:** Modeling of the cerebellar vascular architecture in lobule VI at postnatal days P4, P8, P12, and P21. The sagittal location of the 55- μ m-thick sections used for modeling is indicated in small insets. The interlobular meningeal vessels are shown in purple; the newly formed vessels originating from the interlobular region are shown in blue; the superficial vessels formed within the cerebellar lobule are shown in gray

differentially regulated in at least one stage. The exceptions being *Colla1* and *Timp1* which did not vary significantly across our experiments, and as such, are not represented in Fig. 9.

At stage P4, 30.4% of the genes were regulated (7/23 genes), all of which are upregulated in the IH condition. Most notably, *Vegfa* and two VEGF receptors *Flk1* and *Flt1* are overexpressed, indicating the importance of this pro-angiogenic signaling pathway at this stage. Interestingly, some negative regulators of angiogenesis are also overexpressed, such as endothelial adhesion molecule thrombospondin 1 (*Thbs1*).

P8 is the most regulated stage overall with 73.9% (17/23) differentially expressed genes, but in contrast with P4, all were downregulated under IH. This includes the previously P4 upregulated growth factors but also nearly all genes we tested from the angiopoietin and VEGF signaling pathways. In parallel to this effect on pro-angiogenic molecules, IH also acts negatively on some anti-angiogenic factors such as *Thbs1* and chondromodulin1 (*Lect1*). Moreover, despite being 6 days into the hypoxia protocol, genes involved in hypoxia induced angiogenesis, such as *Vegfa* and the angiopoietin 1 inhibitor *Tie1*, are also downregulated at P8.

Then, P12 had 60.9% (14/23) of regulated genes with a partial switch to upregulation compared to P8. This is the case of the growth factor genes *Fgf2*, *Pgf* and *Tgfb1* but also the anti-angiogenic *Thbs1*. Interestingly, the results showed that IH upregulates the expression of matrix metalloproteinases 2 and 9 involved in vessel invasion and sprouting.

The decrease in IH-regulated genes continues at P21 with only 39.1% (9/23) differentially expressed genes. All of them are under-expressed, including *Flt1*, a negative regulator of neovascularization. Surprisingly, 43.5% (10/23) were still regulated at P70, suggesting that the vascular response to hypoxia is long lasting on a transcriptomic level. This long-term effect decreases pro-angiogenic mechanisms, especially with the downregulation of hypoxia-induced angiogenic factors and positive regulators of vascular invasion and sprouting.

Discussion

An Optimized Method to Visualize the Cerebellar Vascular Network in 3D

Despite the advancements in the study of cerebellar neurogenesis, we still lack knowledge regarding its angiogenesis during development. This lack of information is mainly due to the difficulties in accessing the cerebellar depths with conventional imaging techniques. Fortunately, the advancement of clearing protocols and light sheet microscopy in the past decade have gradually enabled the 3D visualization of the vasculature of the cerebellum in its earliest stages as well as in later ages [38, 39]. However, the analysis of these 3D images is still difficult because only a few software programs can model and produce quantitative data from such 3D files.

One of the typically used software packages for 3D modelling is Imaris from Oxford Instruments Company. However, due to the complexity and the extreme variability of cerebellar vasculature, the IA module is not efficient to model the whole network at once. Therefore, a threshold of intensity needs to first be applied to separate vessels into a superficial network containing the large vessels (>20- μ m diameter) and a deep network consisting of the small deep capillaries. Then machine learning can be used to optimize modelling, but these learning steps must be limited because errors quickly increase due to the accumulation of conflicting information. Furthermore, although this software is the most efficient for modelling, its statistical tools are not adapted to the study of vascular networks. It is therefore necessary to turn to other software solutions to obtain quantitative data. Vessellio [31] proved to be a good solution, as each vascular parameter is well defined. Furthermore, the software is open access and has already been used in other vascular studies. Therefore, after numerous steps of optimization and software tests, a practical, optimal, and user-friendly workflow was created for 3D vascularization analysis.

Developmental Evolution of the Cerebellar Vascularization

Analysis of vascularization during postnatal development shows that the vascular network evolves in parallel with the histogenesis of the cerebellum. Indeed, blood vessel density and network complexity decrease between P4 and P12, then increase at P21. This observation may be related to the fact that the cerebellum undergoes a drastic increase in volume between P4 and P14 [40], which angiogenic activity only compensates for later on.

P4

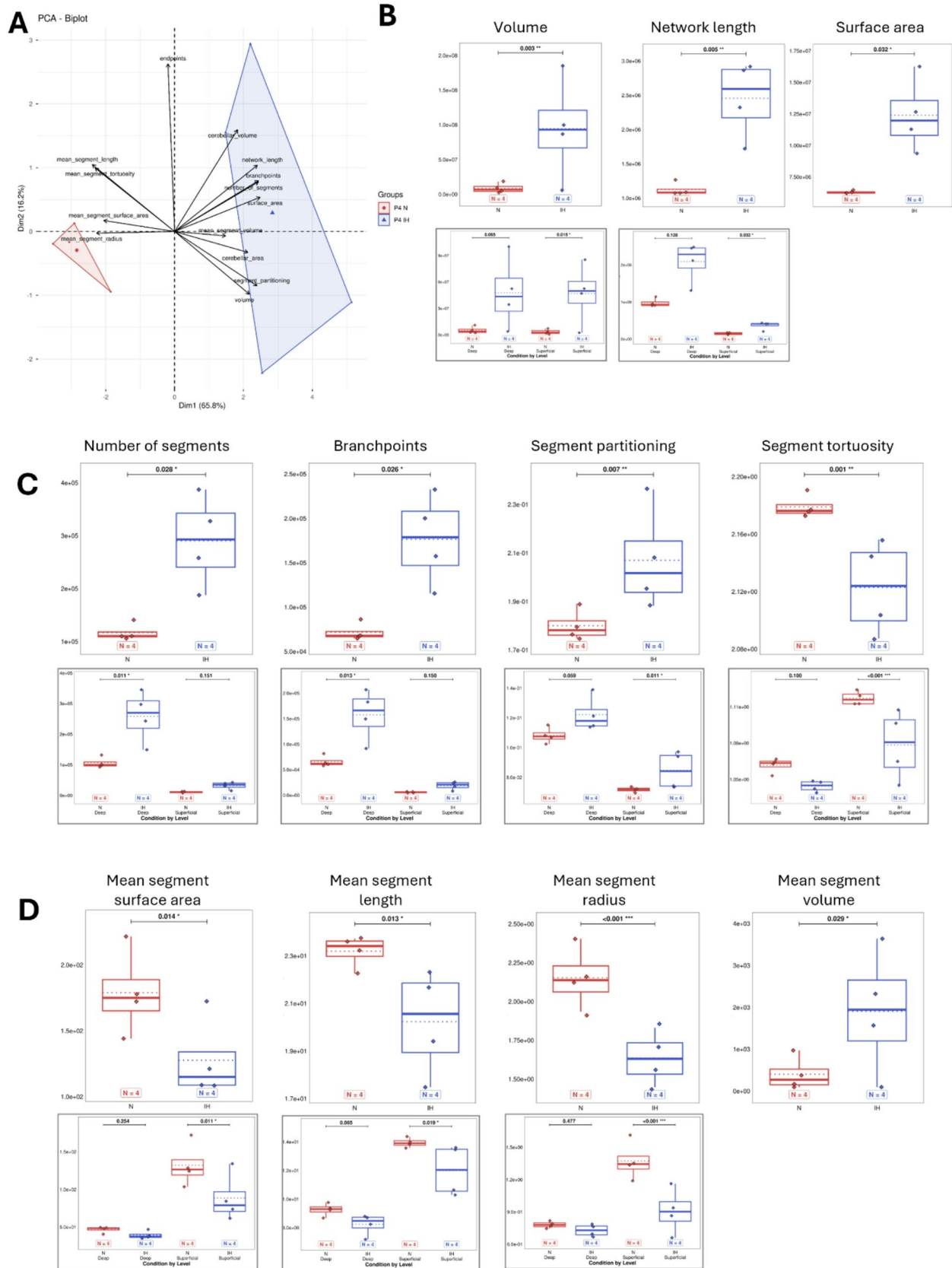


Fig. 5 (next page): Effect of intermittent hypoxia on cerebellar vascularization at an early developmental stage (P4) in mouse. **A**: Projection of samples onto the factorial plane obtained from the principal component analysis (PCA) performed on 14 vascular quantitative variables to illustrate the overall effect of IH on cerebellar angiogenesis at P4 in mice. Each point represents an individual sample, colored according to experimental group: control (red) or hypoxic (blue). **B–D**: Quantification of global cerebellar vascular parameters (**B**), vascular segment parameters (**C**), and mean segment parameters (**D**) in control (red) and hypoxic (blue) mice at P4. Only parameters significantly altered by hypoxia are shown in the figure. The other parameters are available in the supplementary data (suppl Figures 1 to 6). When significant differences could be attributed preferentially to either the deep (left) or superficial (right) network, an additional box plot (outlined in gray) was added below the relevant parameter. For each variable, the total number of animals per experimental group is indicated below the boxplots and represented by diamonds. Exact p-values are displayed above the plots when statistically significant. IH: intermittent hypoxia; N: Normoxia (control); Px: postnatal day x

3D models of physiological cerebellar development show that perforating vessels from the superficial network enter the cerebellum via the cortex and then develop collaterals to supply the entire structure. An intracortical organization is easily recognizable at the level of the convolutions delimiting the lobules. This organization depends on the stage of development of the cerebellum. Thus, at P4, the vessels are few, sparsely branched, and oriented perpendicular to the axis of the lobule. This low density could promote the proliferation of GCP by inducing physiological hypoxia and the expression of HIF1 α . Indeed, it has been shown that this factor maintains granular cells in a proliferative state [41]. This hypothesis is reinforced by the observation of an intermediate plexus that forms at P8 between the EGL and the ML. This plexus would promote oxygenation of the post-mitotic layer of the EGL, thereby allowing the precursors to exit the cell cycle. This new vascularization may also enable the development of the dendritic tree of Purkinje cells, as it has been shown that hypoxia induces a delay in their arborization [7, 42]. At P12, the intermediate plexus shrinks as the EGL gradually disappears, until homogeneous vascularization of the entire cerebellum is achieved at P21 [10, 11].

Early Changes

Our innovative 3D image analysis approach allowed the assessment of the evolution of several vascular parameters in response to our hypoxia protocol. The most changes in the cerebellar vascular network features appear at the P4 stage, after only two days of hypoxia. Overall, the network of IH mice has a higher volume, length and surface area than the control animals mostly attributable to the superficial network. This increase is consistent with the upregulation of several factors promoting vessel growth such as VegfA [43] and its VEGF receptor Flk1 [44, 45], transforming growth factor beta 1 (Tgfb1; [43]), and coagulation factor III (F3;

[46]). This suggests that an acute response to hypoxia at this early stage involves grossly increasing vascularization to compensate for IH exposure. In contrast, the decreased tortuosity of vessels in IH mice, especially in the superficial network, could suggest less remodeling. This hypothesis can be supported by the overexpression of two negative regulators of vascular invasion and sprouting (VIS), namely Flt1 [47, 48] and Thbs1 [49].

The P4 enlarged network is composed of more numerous segments with more branchpoints, especially in the deep network, accompanied by an increased segment partitioning, predominantly in the superficial network. However, the vascular segments of IH mice are shorter and thinner, predominantly in the superficial network. This observation supports the early emphasis on increasing the network rather than refining it after hypoxia, and suggests that hypoxia-induced angiogenesis (HIA) leads to a more complex but less mature vascular network. This hypothesis also correlates with the high upregulation of VegfA which is a known actor of HIA [47] as well as of Angpt2 which promotes EEP in response to hypoxia [50, 51] and of F3 which stimulates neovascularization [46, 52]. Notably, the pro-angiogenic effect of hypoxia has already been demonstrated in pathologies such as cancer, in which tumor cells, via the Warburg effect, modify their metabolism in order to activate endothelial cells and neovascularization [53]. Here, we can postulate a different mechanism involving glial cells and/or mural cells in the pro-angiogenic response. Indeed, astrocytes and pericytes are known to release pro-angiogenic factors such as VEGF or Ang2, which act on tip cells. Furthermore, it has been shown that hypoxia induces astrocyte proliferation and increases their secretory activity, likely making them key players in this response [54, 55].

Vascular Adaptation During IH

In contrast to P4, no significant difference in general network parameters was observed at P8, and only the increase in the mean segment volume is maintained after 6 days of IH. In parallel, all tested genes were downregulated in all functional groups, including the previously upregulated growth factors, but also a large panel of positive regulators of angiogenesis. Among them, we can notice the strong under-expression of the angiogenic receptor Tek and its regulator Tie1 [56, 57], of metalloprotease 2 (MMP2; [58, 59]), of alanyl aminopeptidase (Anpep; [60]), and of Serpine 1 [61], which are all known to be implicated in HIA. This result suggests that the initial acute angiogenesis triggered at the beginning of the protocol in response to IH has ceased by P8. In accordance with this hypothesis, our differential study across networks reveals that the length and the surface area of the superficial vessels increase along with

P8

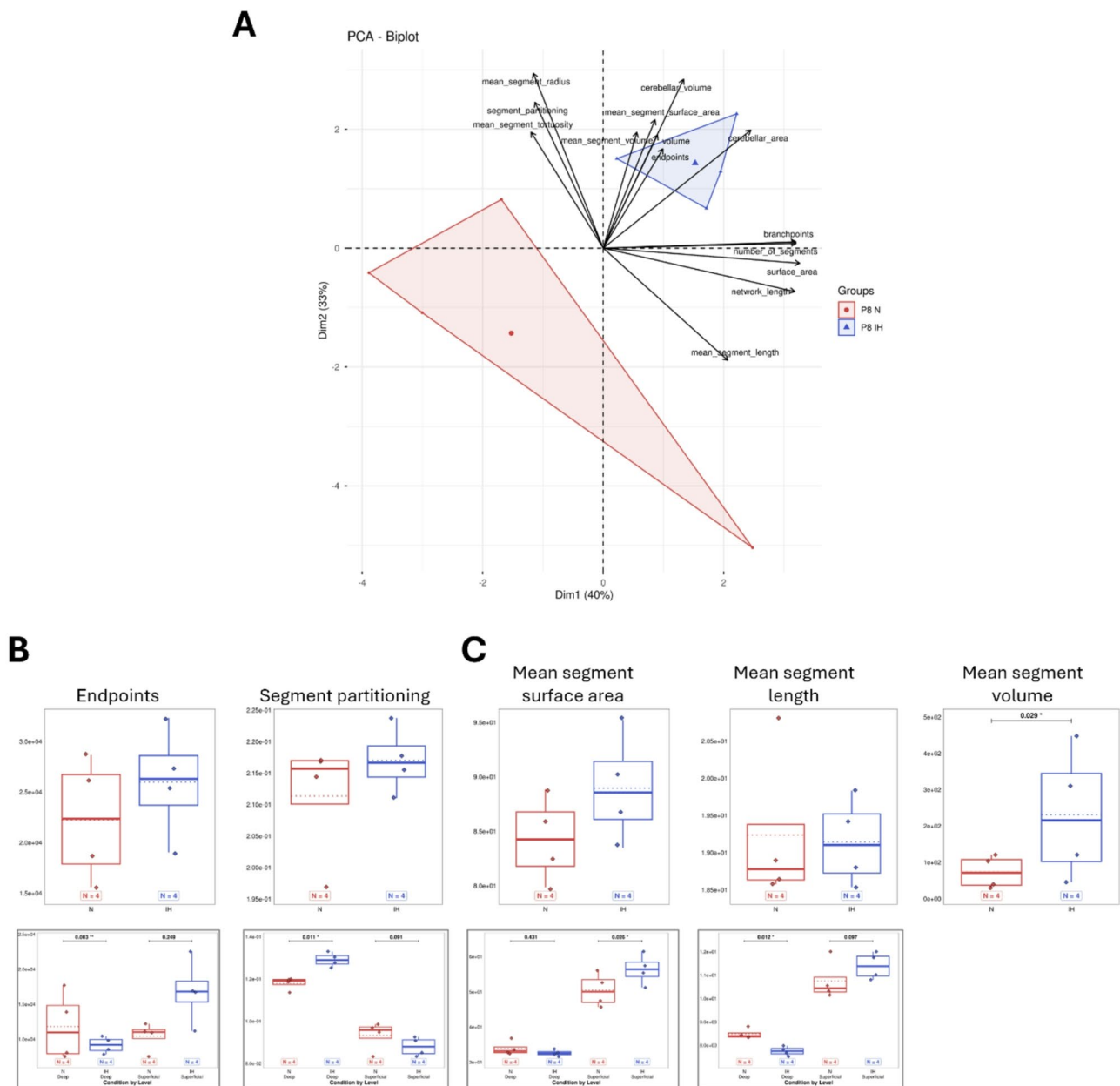


Fig. 6 Effect of intermittent hypoxia on cerebellar vascularization at the postnatal day 8 (P8) in mouse. **A**: Projection of samples onto the factorial plane obtained from the principal component analysis (PCA) performed on 14 vascular quantitative variables to illustrate the overall effect of IH on cerebellar angiogenesis at P8 in mice. Each point represents an individual sample, colored according to experimental group: control (red) or hypoxic (blue). **B–C**: Quantification of vascular segment parameters (**B**), and mean segment parameters (**C**) in control (red) and hypoxic (blue) mice at P8. Only parameters significantly

altered by hypoxia are shown in the figure. The other parameters are available in the supplementary data (suppl Figures 1 to 6). When significant differences could be attributed preferentially to either the deep (left) or superficial (right) network, an additional box plot was added below the relevant parameter. For each variable, the total number of animals per experimental group is indicated below the boxplots and represented by diamonds. Exact p-values are displayed above the plots when statistically significant. IH: intermittent hypoxia; N: Normoxia (control); Px: postnatal day x

P12

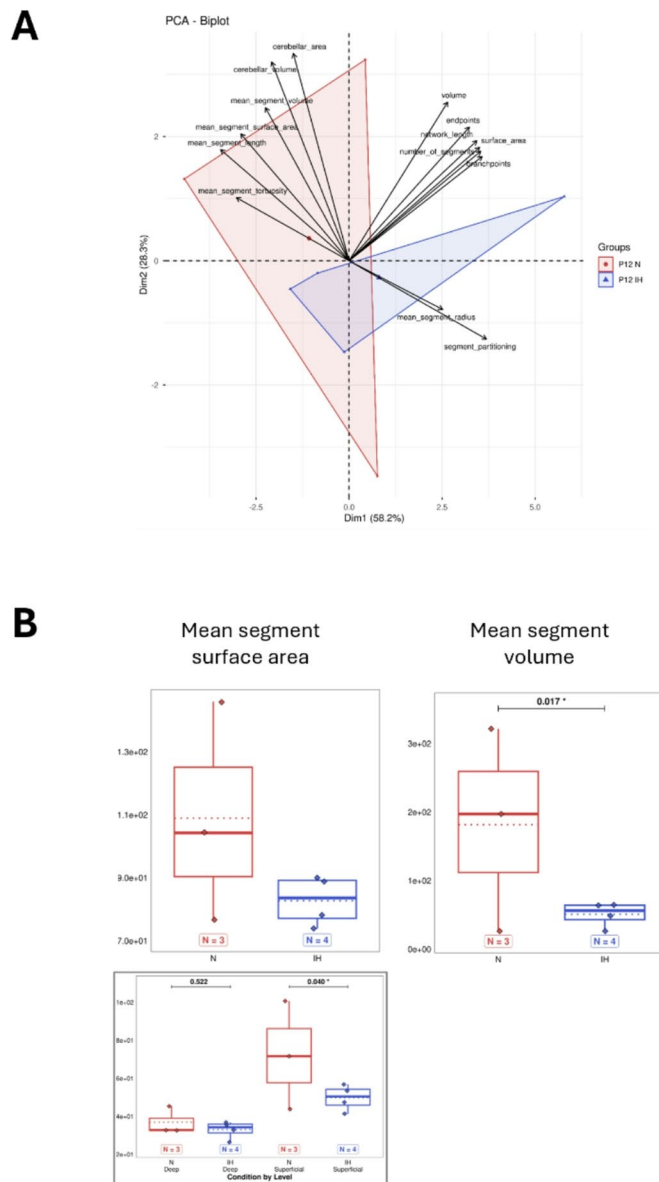


Fig. 7 Effect of intermittent hypoxia on cerebellar vascularization at the postnatal day 12 (P12) in mouse. **A:** Projection of samples onto the factorial plane obtained from the principal component analysis (PCA) performed on 14 vascular quantitative variables to illustrate the overall effect of IH on cerebellar angiogenesis at P12 in mice. Each point represents an individual sample, colored according to experimental group: control (red) or hypoxic (blue). **B:** Quantification of mean segment parameters in control (red) and hypoxic (blue) mice at P12. Only parameters significantly altered by hypoxia are shown in the figure.

a slight decrease of the partitioning. This indicates that the network is still growing but is less complex with fewer vessel interconnections. In contrast, a remodelling process is ongoing in the depth of the cerebellum as an increase of the vessel partitioning associated with a decrease of the number

The other parameters are available in the supplementary data (suppl Figures 1 to 6). When significant differences could be attributed preferentially to either the deep (left) or superficial (right) network, an additional box plot was added below the relevant parameter. For each variable, the total number of animals per experimental group is indicated below the boxplots and represented by diamonds. Exact p-values are displayed above the plots when statistically significant. IH: intermittent hypoxia; N: Normoxia (control); Px: postnatal day x

of endpoints and of the mean segment length are observed. This could be linked to the under-expression of some negative regulators of VIS, namely *Flt1* and *Thbs1* [47–49] or of endothelial cell mechanisms such as *Serpinf1* [62, 63] or *Angpt1* (angiopoietin 1; [50, 64]).

P21

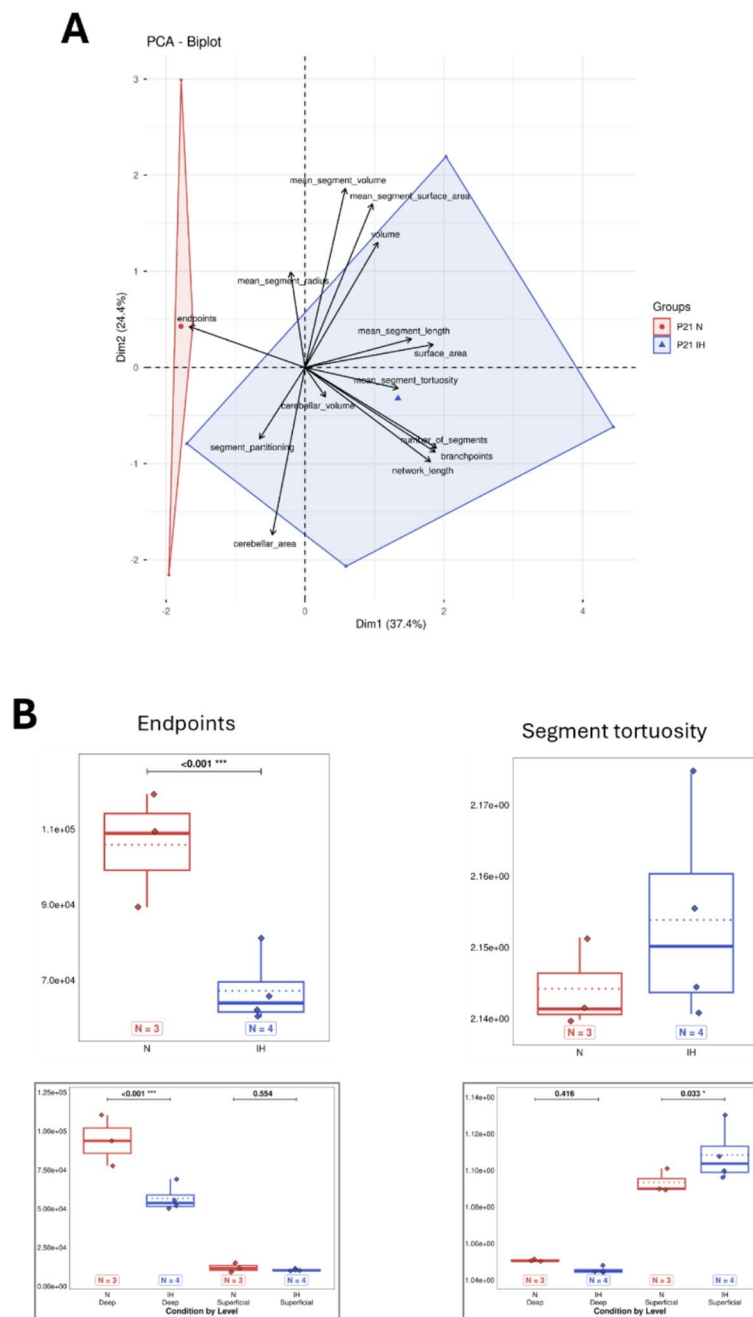


Fig. 8 Effect of intermittent hypoxia on cerebellar vascularization at last developmental stage (P21) in mouse. **A:** Projection of samples onto the factorial plane obtained from the principal component analysis (PCA) performed on 14 vascular quantitative variables to illustrate the overall effect of IH on cerebellar angiogenesis at P21 in mice. Each point represents an individual sample, colored according to experimental group: control (red) or hypoxic (blue). **B:** Quantification of vascular segment parameters in control (red) and hypoxic (blue) mice at P21. Only parameters significantly altered by hypoxia are shown

in the figure. The other parameters are available in the supplementary data (suppl Figures 1 to 6). Since significant differences could be attributed preferentially to either the deep (left) or superficial (right) network, an additional box plot (outlined in gray) was added below the relevant parameter. For each variable, the total number of animals per experimental group is indicated below the boxplots and represented by diamonds. Exact p-values are displayed above the plots when statistically significant. IH: intermittent hypoxia; N: Normoxia (control); Px: postnatal day x

At P12, like at P8, general parameters are not altered but a decrease of the mean segment volume and area are observed, suggesting that the overall shape of blood vessels stabilizes during the protocol. The decrease of the segment size is primarily attributable to the superficial network and could be explained by increased neovascularization of deep vessels intended to reach the entire internal cerebellum. While this effect is not visible in our deep network histological measurements, it could be explained by a partial switch in the expression of several genes. Indeed, while some pro-angiogenic factors remain under-expressed, others become over-expressed such as the matrix metalloproteases MMP2, MMP9, and Angpt2, which make the extracellular matrix more permissive to vessel endothelial tip cells [51, 58, 59, 65]. In addition, *norrin* (*Ndp*), which was previously not affected by IH, becomes under-expressed, magnifying the effect on the matrix [66–69]. Taken together with the upregulation of *Fgf2*, *Pgf* and *Tgfb1*, involved in EPS and VSM, this reveals the emergence of a second phase of IH response at later stages of development [43, 70, 71].

Ten days after the end of the IH protocol, P21 IH mice present a lower number of vessel endpoints than their normoxic counterparts, largely attributable to the deep network. This coincides with the emergence of newly downregulated VIS factors such as the critical Angiopoietin 2 (*Angpt2*; [72] but also *F3* [46, 52], *Fgf2* [73], as well as *Cdh5* [74–76] at P70. At P21, the cerebellum is also characterized by a higher tortuosity of superficial vascular segments, indicating that, despite compensatory mechanisms occurring at the end of the IH protocol, minor alterations to the vascular network remain visible in the long term. Consistent with this observation, the majority of genes under-expressed at P21 are still under-expressed at P70, including *Flk1*, *Tek*, and *Vegfa* [47, 64].

Vasculogenesis and Cerebellar Development During IH

It has been well established that the hypoxic environment experienced by the fetus during gestation is essential for the proper formation of several brain structures. In particular, hypoxia induces the expression of HIF-1 α , a key regulator of fetal vascularization [77] and neuronal survival [78]. However, when this physiological hypoxia is exacerbated, as observed in infant apnea, it becomes deleterious and the cerebellum is not spared. Accordingly, murine models of perinatal apnea exhibit cerebellar hypomyelination, disorganization of cortical layers, and Purkinje cell alterations associated with behavioral deficits [6, 7, 42].

The present work suggests that these effects may be partly mediated by impaired hypoxia-induced angiogenesis. Indeed, the vascular densification observed at P4 is not maintained, and may even decrease over time, potentially maintaining

granule cell precursors in a proliferative state [41]. This could explain the increased proliferation and the persistence of a prominent external granular layer (EGL) at P12 in our AOP model [7]. Likewise, since the dendritic arborization of Purkinje cells only intensifies from P6 onward, it may not benefit from the early hypoxia-induced neo-angiogenesis and could therefore be compromised [7, 42]. Our previous findings also indicate that Purkinje cell sensitivity to hypoxia varies along the anteroposterior axis. At P12, alterations are predominantly observed in the anterior regions of the cerebellum, whereas compensatory mechanisms first arise in posterior regions before progressively extending to the entire cerebellum [7]. This process could be explained by the developmental timing of cerebellar vascularization. Indeed, the superior cerebellar artery (SCA), which supplies the anterior cerebellum, develops earlier than the anterior inferior cerebellar artery (AICA) and the posterior inferior cerebellar artery (PICA) during embryonic development [79]. These observations suggest that the vascular network does not reach the same degree of maturation simultaneously across cerebellar territories, thereby conferring a differential neuronal sensitivity to perinatal hypoxia.

Beyond its classical role as a supplier of O₂, the vascular system is now recognized as a critical regulator of neural development. Interactions between blood vessels and neural cells are essential for several cellular processes occurring during brain development, including precursor proliferation, migration, and differentiation [80]. Because cerebellar development occurs predominantly during the postnatal period in both mice and humans, vascular abnormalities induced by our intermittent hypoxia (IH) protocol—such as reduced vessel tortuosity or diameter—may contribute to the cortical layer disorganization we previously observed at P12 [7].

In addition, neurons and endothelial cells share a wide range of signaling pathways [81, 82], raising the possibility that both the nervous and vascular systems are affected in parallel by IH. For instance, matrix metalloproteinases are involved not only in cerebral angiogenesis [83] but also in cerebellar cortical morphogenesis [84]. Regarding migratory processes, neuropilin-1 (NRP1) acts as a guidance factor for both neurons and blood vessels [85, 86], through its interaction with either VEGFA [87] or SEMA3A [88]. Similarly, VEGF produced in the cerebellum promotes granule cell migration through the molecular layer via the FLK1 receptor [89]. Another example is the Ang2/Tie2 signaling axis, well known for its role in vascular development [17], which has more recently been identified as a regulator of dendritic tree development in Purkinje cells [18]. Since each of these pathways is altered by our IH protocol, their dysregulation may account for the concomitant emergence of vascular defects and neuronal alterations observed at P12, including disorganization of cerebellar cortical layers and

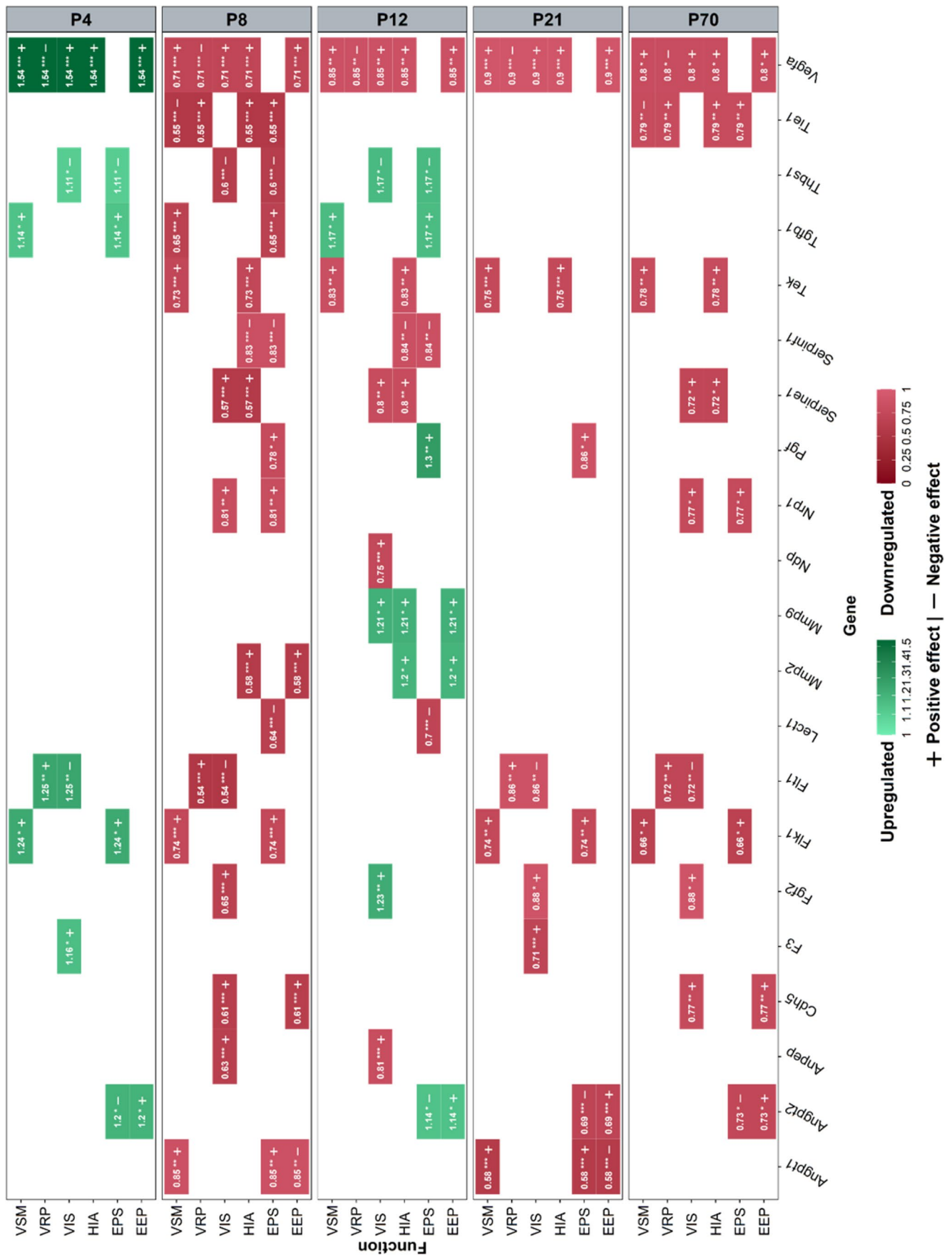


Fig. 9 (next page): Effect of intermittent hypoxia on angiogenesis-related genes during cerebellar development. Significant results reflecting the regulation of the expression of genes involved in cerebellar angiogenesis after IH in P4, P8, P12, P21, and P70 mice (right y-axis) measured by qRT-PCR. Upregulated (green) and downregulated (red) genes are represented for each pathway and the color corresponds to the gradient of the fold change value indicated in each box. Each gene (x-axis) can be mapped back to one or several functional pathways (left y-axis). EEP: extracellular matrix and endothelial permeability; EPS: endothelial proliferation and survival; HIA: hypoxia-induced angiogenesis; Px: postnatal day x; VIS: vascular invasion and sprouting; VRP: vascular remodeling and patterning; VSM: vascular stabilization and maturation. Asterisks indicate the level of statistical significance: one for $p < 0.05$, two for $p < 0.01$, and three for $p < 0.001$.

Purkinje dendritic atrophy [7]. Finally, although these histological abnormalities tend to diminish from P21 onward, the expression of numerous angiogenesis-related genes remains dysregulated and may contribute to the behavioral impairments observed in adulthood following IH exposure [7, 8].

Conclusion

AOP corresponds to breathing pauses, mainly linked to immaturity of the respiratory centers, associated with neurodevelopmental disorders in the short and long terms. In fact, current treatments mainly aim to stimulate breathing and manage the neurological consequences when they appear. The current standard treatment for AOP is therefore the use of methylxanthines such as caffeine or aminophylline, which act as inhibitors of adenosine A1 and A2 receptors, or molecules that stimulate carotid chemoreceptors, such as doxapram or ENA-001 [90, 91]. However, this work, together with our previous studies, demonstrates that neurogenesis and angiogenesis are tightly interlocked during cerebellar postnatal development and that both systems are affected by a perinatal IH. Thus, the two components participate in deficits observed in patients and it would be a mistake to focus therapeutic approaches solely on neurological damage, without considering the vascular aspect. Therefore, treatments combining pro-angiogenic molecules with respiratory stimulants, such as the erythropoietin/caffeine combination tested in rodents [92], could be promising therapeutic approaches.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12311-026-02006-1>.

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Author Contributions A.R.D. participated in Investigation, Methodology, Project administration, Data Curation, Formal analysis, Visualization, Validation, Writing - Original Draft, Writing - Review and Editing. C.R. contributed to Investigation, Writing - Original Draft, Writing - Review and Editing. M.B.D. collaborated on Supervision, Resources, Writing - Review and Editing while D.V. took part in Funding acquisition, Resources, Writing - Review and Editing. B.G. participated in Funding acquisition, Resources, Writing - Review and Editing. D.B. contributed to Investigation, Conceptualization, Methodology, Funding acquisition, Project administration, Supervision, Resources, Writing - Original Draft, Writing - Review and Editing.

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Data Availability The raw datasets, analyses and code supporting this work can be accessed via the GitHub repository available at <https://github.com/agalic-rd/Vasc-AoP>, also referenced on Zenodo under: <https://doi.org/10.5281/zenodo.15319299>.

Declarations

Competing interests The authors declare no competing interests.

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